Descriptive Statistics and Introduction to Probability
Problem sheet N°4

Exercice 1. Suppose a family contains two children of different ages, and we are interested in the gender of these children. Let F denote that a child is female and M that the child is male and let a pair such as FM denote that the older child is female and the younger is male. There are four points in the set Ω of possible observations: $\Omega = \{\text{FF, FM, MF, MM}\}$. Let A denote the subset of possibilities containing no males; B, the subset containing two males; and C, the subset containing at least one male.
List the elements of : $A, B, C, A \cap B, A \cup B, A \cap C, A \cup C, B \cap C, B \cup C$ and $C \cap B^c$.
Exercice 2. Use the identities $A = A \cap \Omega$ and $B = B \cap \Omega$ and a distributive law to prove that

1. $A = (A \cap B) \cup (A \cap B^c)$.
2. If $B \subset A$ then $A = B \cup (A \cap B^c)$.
3. Further, show that $(A \cap B)$ and $(A \cap B^c)$ are mutually exclusive and therefore that A is the union of two mutually exclusive sets.
4. Also, show that B and $(A \cap B^c)$ are mutually exclusive and if $B \subset A$, A is the union of two mutually exclusive sets B and $(A \cap B^c)$.

Exercice 3. A sample space consists of five simple events, E_1, E_2, E_3, E_4, and E_5.

1. If $P(E_1) = P(E_2) = 0.15$, $P(E_3) = 0.4$ and $P(E_4) = 2P(E_5)$, find the probabilities of E_4 and E_5.
2. If $P(E_1) = 3P(E_2) = 0.3$, find the probabilities of the remaining simple events if you know that the remaining simple events are equally probable.

Exercice 4. A local fraternity is conducting a raffle where 50 tickets are to be sold one per customer. There are three prizes to be awarded. If the four organizers of the raffle each buy one ticket, what is the probability that the four organizers win

1. all of the prizes?
2. exactly two of the prizes?
3. exactly one of the prizes?
4. none of the prizes?

Exercice 5. 1. There are total 18 balls in a bag. Out of them 6 are red in colour, 4 are green in colour and 8 are blue in colour. If someone picks three balls randomly from the bag, then what will be the probability that all the three balls are not of the same colour?
2. Bag A contains 3 green and 7 blue balls. While bag B contains 10 green and 5 blue balls. If one ball is drawn from each bag, what is the probability that both are green?
Exercice 6. A study is to be conducted in a hospital to determine the attitudes of nurses toward various administrative procedures. A sample of 10 nurses is to be selected from a total of the 90 nurses employed by the hospital.

1. How many different samples of 10 nurses can be selected?

2. Twenty of the 90 nurses are male. If 10 nurses are randomly selected from those employed by the hospital, what is the probability that the sample of ten will include exactly 4 male (and 6 female) nurses?

Exercice 7. Five firms, F1, F2, . . . , F5, each offer bids on three separate contracts, C1,C2, and C3. Any one firm will be awarded at most one contract. The contracts are quite different, so an assignment of C1 to F1, say, is to be distinguished from an assignment of C2 to F1.

1. How many sample points are there altogether in this experiment involving assignment of contracts to the firms? (No need to list them all.)

2. Under the assumption of equally likely sample points, find the probability that F3 is awarded a contract.

Exercice 8. Suppose that we ask n randomly selected people whether they share your birthday.

1. Give an expression for the probability that no one shares your birthday (ignore leap years).

2. How many people do we need to select so that the probability is at least .5 that at least one shares your birthday?

Exercice 9. A labor dispute has arisen concerning the distribution of 20 laborers to four different construction jobs. The first job (considered to be very undesirable) required 6 laborers; the second, third, and fourth utilized 4, 5, and 5 laborers, respectively. The dispute arose over an alleged random distribution of the laborers to the jobs that placed all 4 members of a particular ethnic group on job 1. In considering whether the assignment represented injustice, a mediation panel desired the probability of the observed event.

1. Determine the number of sample points in the sample space Ω for this experiment. That is, determine the number of ways the 20 laborers can be divided into groups of the appropriate sizes to fill all of the jobs.

2. Find the probability of the observed event if it is assumed that the laborers are randomly assigned to jobs.

Exercice 10. Three brands of coffee, X, Y , and Z, are to be ranked according to taste by a judge. Define the following events: A:" Brand X is preferred to Y", B: "Brand X is ranked best", C:" Brand X is ranked second best" and D: "Brand X is ranked third best".
If the judge actually has no taste preference and randomly assigns ranks to the brands, is event A independent of events B, C, and D?
Exercice 11. If two events, A and B, are such that $P(A) = 0.5, P(B) = 0.3$ and $P(A \cap B) = 0.1$, find the following:

1. $P(A|B)$ and $P(B|A)$

2. $P(A|A \cup B)$ and $P(A|A \cap B)$

3. $P(A \cap B|A \cup B)$

Exercice 12. 1. If A and B are independent events, show that A and B^c are also independent. Are A^c and B^c independent?
2. Can A and B be mutually exclusive if $P(A) = 0.4$ and $P(B) = 0.7$? If $P(A) = 0.4$ and $P(B) = 0.3$? Why?
Exercice 13. It is known that a patient with a disease will respond to treatment with probability equal to 0.9. If three patients with the disease are treated and respond independently, find the probability that at least one will respond.
Exercice 14. An advertising agency notices that approximately 1 in 50 potential buyers of a product sees a given magazine ad, and 1 in 5 sees a corresponding ad on television. One in 100 sees both. One in 3 actually purchases the product after seeing the ad, 1 in 10 without seeing it. What is the probability that a randomly selected potential customer will purchase the product?
Exercice 15. An electronic fuse is produced by five production lines in a manufacturing operation. The fuses are costly, are quite reliable, and are shipped to suppliers in 100-unit lots. Because testing is destructive, most buyers of the fuses test only a small number of fuses before deciding to accept or reject lots of incoming fuses. All five production lines produce fuses at the same rate and normally produce only 2% defective fuses, which are dispersed randomly in the output. Unfortunately, production line 1 suffered mechanical difficulty and produced 5% defectives during the month of March. This situation became known to the manufacturer after the fuses had been shipped. A customer received a lot produced in March and tested three fuses. One failed.

1. What is the probability that the lot was produced on line 1?
2. What is the probability that the lot came from one of the four other lines?

Exercice 16. A diagnostic test for a disease is such that it (correctly) detects the disease in 90% of the individuals who actually have the disease. Also, if a person does not have the disease, the test will report that he or she does not have it with probability .9. Only 1% of the population has the disease in question. If a person is chosen at random from the population and the diagnostic test indicates that she has the disease, what is the conditional probability that she does, in fact, have the disease? Are you surprised by the answer? Would you call this diagnostic test reliable?
Exercice 17. A student answers a multiple-choice examination question that offers four possible answers. Suppose the probability that the student knows the answer to the question is .8 and the probability that the student will guess is .2. Assume that if the student guesses, the probability of selecting the correct answer is .25. If the student correctly answers a question, what is the probability that the student really knew the correct answer?
Exercice 18. Five identical bowls are labeled $1, 2, 3, 4$ and 5. Bowl i contains i white and $5 - i$ black balls, with $i = 1, 2, ..., 5$. A bowl is randomly selected and two balls are randomly selected (without replacement) from the contents of the bowl.

1. What is the probability that both balls selected are white?
2. Given that both balls selected are white, what is the probability that bowl 3 was selected?

Exercice 19. A fair coin is tossed repeatedly until either it lands heads or a total of five tosses have been made, whichever comes first. Let X denote the number of tosses made.

1. Construct the probability distribution for X.
2. Give the CDF of X.

3. What is the probabilty that X will be greater than 3.

4. Given that a head is obtained, what is the probability that X is less or equal to 3.

Exercice 20. A discrete random variable X has the following probability distribution :

x_i	77	78	79	80	81
$P(X = x_i)$	0.15	0.15	0.2	0.4	0.1

Compute each of the following quantities.

1. F the CDF of X.

2. $P(X = 80), P(X > 80)$ and $P(X \leq 80)$.

3. The mean μ_X, the variance σ_X^2 and the standard deviation σ_X.

Exercice 21. Two fair dice are rolled at once. Let X denote the difference in the number of dots that appear on the top faces of the two dice. Thus for example if a one and a five are rolled, $X = 4$, and if two sixes are rolled, $X = 0$.

1. Construct the probability distribution for X.

2. Give the CDF of X.

3. Compute the mean and the standard deviation of X.